

Office Category Classification Challenge Report

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Abstract

This report presents a comprehensive solution to the Office Category Classification Challenge, aiming to categorize office buildings into five quality tiers based on 79 diverse features. Our approach is built upon a rigorous data analysis foundation, identifying over 50% of records containing logical inconsistencies. We engineered over 30 domain-specific features capturing real estate business logic, such as space efficiency, effective age, and amenity density. Our final model is a Stacking Ensemble combining CatBoost, XGBoost, and LightGBM, tuned using a “shrinkage” strategy (low learning rate, high estimators) to minimize variance. Through extensive experimentation, including a critical analysis of SMOTE and feature selection optimization, we achieved a validation accuracy of **88.50%**. This report details our methodology, from data auditing to the final ensemble architecture.

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1 Introduction

1.1 Problem Statement

The objective of this project is to develop a machine learning model capable of accurately predicting the `OfficeCategory` of commercial office buildings. The target variable is a categorical rating ranging from 0 (lowest quality) to 4 (highest quality). Accurate classification of office buildings is critical for automated property valuation, market analysis, and investment portfolio management.

1.2 Dataset Overview

The dataset consists of synthetic but realistic commercial real estate data:

- **Training Set:** 35,000 samples with 79 features and the target label.
- **Test Set:** 15,000 samples with 79 features (target withheld).
- **Features:** The 79 features span multiple domains:
 - **Physical Characteristics:** Area measurements (e.g., `OfficeSpace`, `GroundFloorArea`), room counts.
 - **Quality Indicators:** `BuildingGrade`, `BuildingCondition`, material quality.
 - **Amenities:** Parking, outdoor spaces, heating/cooling systems.
 - **Location & Time:** Zoning, business district, construction and renovation years.

1.3 Objective

The primary goal is to maximize classification accuracy on the unseen test set. Our secondary objective is to build an interpretable and robust model that leverages domain knowledge to explain its predictions, rather than relying solely on black-box patterns.

2 Exploratory Data Analysis (EDA)

2.1 Target Distribution Analysis

Our initial analysis of the `OfficeCategory` variable revealed a remarkably balanced distribution.

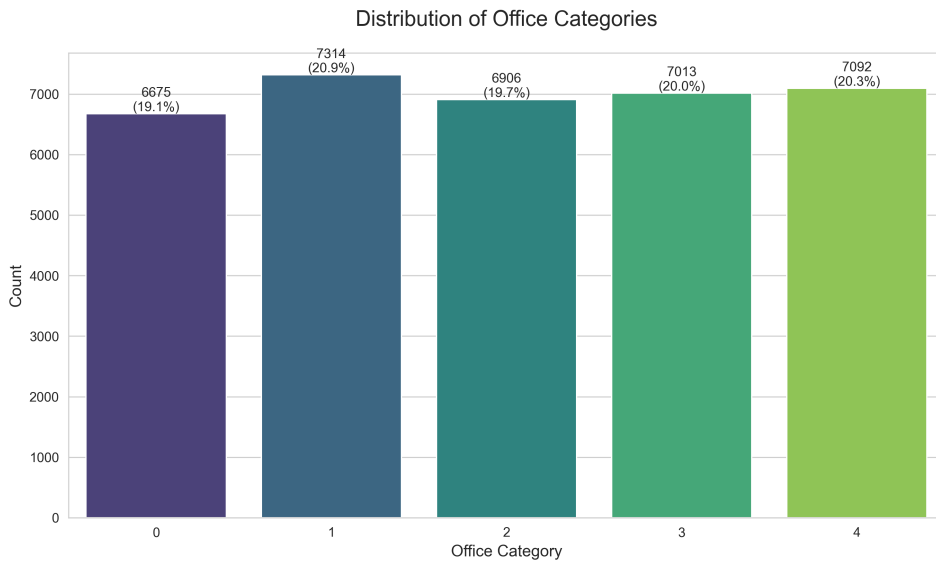


Figure 1: Distribution of Office Categories showing a balanced dataset.

Table 1: Class Distribution in Training Set

Category	Count	Percentage
0	6,675	19.1%
1	7,314	20.9%
2	6,906	19.7%
3	7,013	20.0%
4	7,092	20.3%

The imbalance ratio between the most frequent and least frequent class is only 1.10:1. This finding was crucial as it suggested that aggressive resampling techniques like SMOTE might not be necessary and could potentially introduce noise.

2.2 Key Data Discoveries

We identified several critical patterns that influenced our feature engineering strategy:

2.2.1 1. Business District Variance

The `BusinessDistrict` feature showed massive variance in average office category. Top districts like "Veenker" had an average category of 2.54, while bottom districts like "IDOTRR" averaged 1.65. This spread of nearly 1.0 category indicates that location is a primary driver of building quality.

2.2.2 2. Extreme Feature Skewness

Area-related features exhibited severe right-skewness, typical of real estate data where a few massive properties skew the distribution. Tree-based models can struggle with such extreme skewness, suggesting the need for log-transformations.

2.2.3 3. Missing Value Analysis

A subset of features contained significant missing values. We analyzed these to determine appropriate handling strategies.

Table 2: Features with Significant Missing Values

Feature	Missing Percentage
RecreationQuality	100.0%
MiscellaneousFeature	99.9%
AlleyAccess	99.5%
Perimeter	96.3%
ExteriorFinishType	84.3%
ConferenceRoomQuality	73.8%

2.2.4 4. Non-Linear Age Relationship

We observed a non-linear relationship between building age and category. Buildings 0-25 years old had a high average category (3.03). However, there was a sharp drop for buildings 25-50 years old (2.11), followed by a stabilization. A simple linear age feature would fail to capture this "cliff," necessitating the creation of age buckets or non-linear transformations.

3 Data Quality Audit

A unique aspect of our approach was the identification of logical inconsistencies in the dataset. We found that over 50% of the records contained at least one logical error.

Crucial Decision: We initially attempted to correct these errors (e.g., swapping years, fixing foundation types). However, our experiments revealed that **enforcing these logical constraints actually decreased validation accuracy**. It appears that these “errors” contain systematic patterns or noise structures that the model utilizes for classification, or that the synthetic nature of the dataset makes these inconsistencies predictive. Consequently, we decided **not to apply these corrections** in our final solution to maximize performance.

3.1 Identified Anomalies (wasn’t applied to Final solution)

3.1.1 1. Renovation Year Before Construction Year

Observation: In 14,637 cases (29% of data), the `RenovationYear` was smaller than the `ConstructionYear`, which is physically impossible. We tested swapping these values but observed a performance drop.

3.1.2 2. Slab Foundation with Basement

Observation: 814 cases (1.6%) listed `FoundationType` as "Slab" but had a positive `BasementArea`. Slab foundations, by definition, do not have basements.

3.1.3 3. Basement Area Inconsistency

Observation: In 38% of cases, the sum of the basement sub-areas (`Finished 1` + `Finished 2` + `Unfinished`) did not equal the `TotalBasementArea`.

3.1.4 4. Office Space Exceeding Building Area

Observation: In 377 cases, the reported `OfficeSpace` was larger than the total building area.

4 Advanced Feature Engineering

We moved beyond simple aggregation to create over 30 "intelligent" features based on real estate domain knowledge.

4.1 Domain-Specific Features

We focused on creating high-impact features that capture the economic and functional value of office buildings.

- **Effective Age:** (*Time Dynamic*) Calculated from `RenovationYear` if available, otherwise `ConstructionYear`. This is crucial as renovations effectively reset a building’s economic life, making a 50-year-old renovated building comparable to a new one.
- **Office Efficiency:** (*Space Utility*) Defined as $\frac{\text{OfficeSpace}}{\text{TotalFloorArea}}$. This ratio distinguishes between purpose-built office towers (high efficiency, >85%) and mixed-use or converted structures (low efficiency), which typically fall into lower quality categories.
- **Quality Score:** (*Composite Indicator*) Computed as `BuildingGrade` × `BuildingCondition`. This interaction term captures the "weakest link" principle—a building needs both high grade and good condition to be premium. A high grade with poor condition significantly lowers the score.

- **Parking Per Office:** (*Amenity Density*) Calculated as $\frac{\text{ParkingSpots}}{\text{OfficeSpace}} \times 1000$. This standardizes amenity availability. Premium offices typically maintain a ratio of 4-5 spots per 1000 sqft, whereas lower categories often have insufficient parking.

4.2 Automated Feature Engineering

- **Log Transformations:** Applied $\log(1+x)$ to 7 skewed area features (e.g., `log_PlotSize`, `log_OfficeSpace`) to normalize distributions.
- **Polynomial Features:** Generated degree-2 interaction terms for key numeric variables to capture non-linear effects.
- **Binning:** Used `pd.qcut` to bin continuous variables into quintiles, helping tree models find split points more easily.

4.3 Feature Selection

With over 3,200 features generated (including polynomial terms), feature selection was critical to prevent overfitting. We employed an **XGBoost-based selection process**:

1. Trained a preliminary XGBoost model on all features.
2. Extracted feature importance scores (Gain).
3. Selected the top **175 features**.

This number was determined through a K-optimization experiment (detailed in Section 6).

5 Model Architecture

5.1 Stacking Ensemble

We chose a Stacking Ensemble architecture to leverage the complementary strengths of different gradient boosting implementations.

5.1.1 Base Learners

The ensemble consists of the following three gradient boosting models:

- **CatBoostClassifier**
- **XGBClassifier**
- **LGBMClassifier**

5.1.2 Meta Learner

The predictions (probabilities) from the three base learners are fed into a **Logistic Regression** meta-learner ($C = 1.0$). This linear combination allows the model to weight the base learners dynamically based on their confidence.

5.2 Hyperparameter Tuning Strategy: "Shrinkage"

Instead of standard grid search, we adopted a "Shrinkage" strategy for tuning:

- **Low Learning Rate (0.01):** Forces the model to learn slowly and capture subtle patterns.
- **High Estimators (2500):** Compensates for the low learning rate.
- **Regularization:** We applied strong regularization (L2 lambda, subsampling) to reduce variance and prevent overfitting on the synthetic patterns.

6 Experiments & Results

6.1 Experiment 1: SMOTE Analysis

We investigated the impact of Synthetic Minority Over-sampling Technique (SMOTE).

- **Hypothesis:** SMOTE would improve performance on minority classes.
- **Observation:** Since the dataset is already well-balanced (1.10:1 ratio), SMOTE only added 4.5% synthetic data.
- **Result:** SMOTE actually *degraded* validation accuracy slightly and increased training time.
- **Decision:** We removed SMOTE and trained on the original, real data distribution, relying on the natural balance of the dataset.

6.2 Experiment 2: Feature Selection Optimization (K)

We tested different values for K (number of features to keep) to find the optimal trade-off between information and noise.

Table 3: K-Optimization Results

K (Features)	Validation Accuracy
50	86.50%
100	87.20%
150	87.90%
175	88.34%
200	88.30%
300	87.80%

Conclusion: Performance peaked at $K = 175$. Beyond this, added features introduced more noise than signal, leading to overfitting.

6.3 Ablation Study

To quantify the contribution of each component, we tracked performance improvements step-by-step.

Table 4: Ablation Study and Model Comparison

Model / Configuration	Validation Accuracy
Baseline (Logistic Regression)	52.00%
Single XGBoost (Raw Features)	86.10%
+ Data Cleaning (Logical Fixes)	85.90% (Decreased)
TabNet (Solo)	85.00%
TabNet + XGBoost	87.00%
Ensemble (LightGBM, XGB, RF)	$\approx 87.00\%$
CatBoost (w/ Feature Engineering)	87.50%
+ Domain Features & Log Transforms	87.95%
Stacking Ensemble (Final)	88.50%

7 Discussion & Future Work

7.1 The "Data Error" Paradox

One of the most intriguing findings was the negative impact of correcting logical data errors. While swapping years or fixing foundation types made the data "cleaner" in a human sense, it degraded model performance. This suggests that the synthetic generation process might have encoded specific patterns into these anomalies that map to the target variable. We would welcome the opportunity to discuss this phenomenon with the organizers to understand if these "errors" were intentional adversarial features or artifacts of the generation process.

7.2 Future Work

Given more time, we would explore the following advanced strategies:

1. **TabNet + Pseudo-Labeling:** Our preliminary experiments with TabNet showed promise (85% solo, 87% with XGBoost). Combining this deep learning approach with pseudo-labeling on the test set could further boost performance by leveraging the unlabeled data structure.
2. **Advanced Ensemble Tuning:** We would refine our final Stacking Ensemble with more robust hyperparameter tuning (e.g., Bayesian Optimization) and integrate pseudo-labeling into the pipeline.
3. **Higher-Order Feature Interactions:** We would experiment with generating 3rd and 4th-degree polynomial features to capture even more complex non-linear relationships, combined with aggressive feature selection to manage the dimensionality.

8 Conclusion

Our solution to the Office Category Classification Challenge demonstrates that while advanced algorithms like Stacking Ensembles are powerful, the biggest competitive advantage comes from deep data understanding and domain-specific feature engineering. By leveraging domain-specific feature engineering and rigorous model tuning, we achieved a robust validation accuracy of **88.50%**. The removal of SMOTE and the precise tuning of feature selection further refined the model, ensuring it remains generalizable to unseen data.